In [1]:

```python
import numpy as np
from scipy import stats

# self supervised typiclust, original vs modified
budgets = [10, 20, 30, 40, 50, 60]
original_acc = np.array([28.5, 18.0, 22.0, 38.5, 46.5, 42.5])
modified_acc = np.array([36.5, 45.5, 46.0, 50.0, 48.5, 48.0])

mean_orig = np.mean(original_acc)
mean_mod = np.mean(modified_acc)
mean_diff = mean_mod - mean_orig

peak_orig = np.max(original_acc)
peak_mod = np.max(modified_acc)
peak_diff = peak_mod - peak_orig

t_stat, p_value = stats.ttest_rel(modified_acc, original_acc)

print(" Statistical Evaluation (Self-Supervised Embeddings) ")
print(f"Mean Acc. (Original): {mean_orig:.2f}%")
print(f"Mean Acc. (Modified): {mean_mod:.2f}%")
print(f"Difference:           +{mean_diff:.2f}%\n")

print(f"Peak Acc. (Original): {peak_orig:.2f}%")
print(f"Peak Acc. (Modified): {peak_mod:.2f}%")
print(f"Difference:           +{peak_diff:.2f}%\n")

print(f"T-statistic:          {t_stat:.4f}")
print(f"P-value:              {p_value:.4f}")
```

```
 Statistical Evaluation (Self-Supervised Embeddings)
Mean Acc. (Original): 32.67%
Mean Acc. (Modified): 45.75%
Difference:           +13.08%

Peak Acc. (Original): 46.50%
Peak Acc. (Modified): 50.00%
Difference:           +3.50%

T-statistic:          3.0961
P-value:              0.0270
```